

Automated Dog Breed Identification from Shelter Photos

Capstone Project Final Report

1. Problem

Shelter staff assign breed labels to dogs visually at intake, and this practice is known to be inconsistent -- research comparing visual breed calls to DNA-verified breed composition (Voith et al., 2009) found only 25% of adoption agencies' visual breed identifications matched the DNA-identified predominant breed. Inaccurate breed labels carry real consequences: they affect adoption outcomes, breed-restricted housing and insurance eligibility, and skew breed-specific bite-report data. This project builds an image classifier that predicts a dog's breed from a photo, to give shelters a more consistent, evidence-informed alternative to visual guesswork alone.

2. Approach

A MobileNetV2 image classifier (pretrained on ImageNet, via the Hugging Face Model Hub) was fine-tuned on the Stanford Dogs Dataset -- 20,486 images across 115 breed classes (some visually near-identical size/coat variants of the same breed, e.g. Toy/Miniature/Standard Poodle, were deliberately grouped together during data wrangling). The dataset was split 70/15/15 into train/validation/test, stratified by breed. Three fine-tuning strategies were compared to isolate how much backbone adaptation actually helps: a frozen backbone (Model A), the last three backbone blocks unfrozen (Model B), and the full backbone unfrozen (Model C).

A second, independent evaluation set was built from PetFinder.my listings -- real, non-staged shelter photos, as opposed to Stanford Dogs' cleaner, more posed images -- to test whether performance holds up on the kind of photos a real shelter would actually have. Of 7,938 usable PetFinder dog listings, 1,604 had a breed label that mapped onto Stanford Dogs' 115-class vocabulary (harmonized through exact matching, known-synonym mapping for outdated breed names, and token-subset matching, e.g. 'Maltese' to 'Maltese Dog'); the remainder were mostly 'Mixed Breed' listings or breeds outside Stanford Dogs' scope, and were excluded from breed-level evaluation rather than force-matched.

3. Results

Model C (fully unfrozen) was the best performer on every metric, on both evaluation sets, with no sign yet of overfitting from the added capacity. Fine-tuning depth showed a clear, consistent trend: more of the backbone allowed to adapt, the better the result -- at every step from frozen to partial to full.

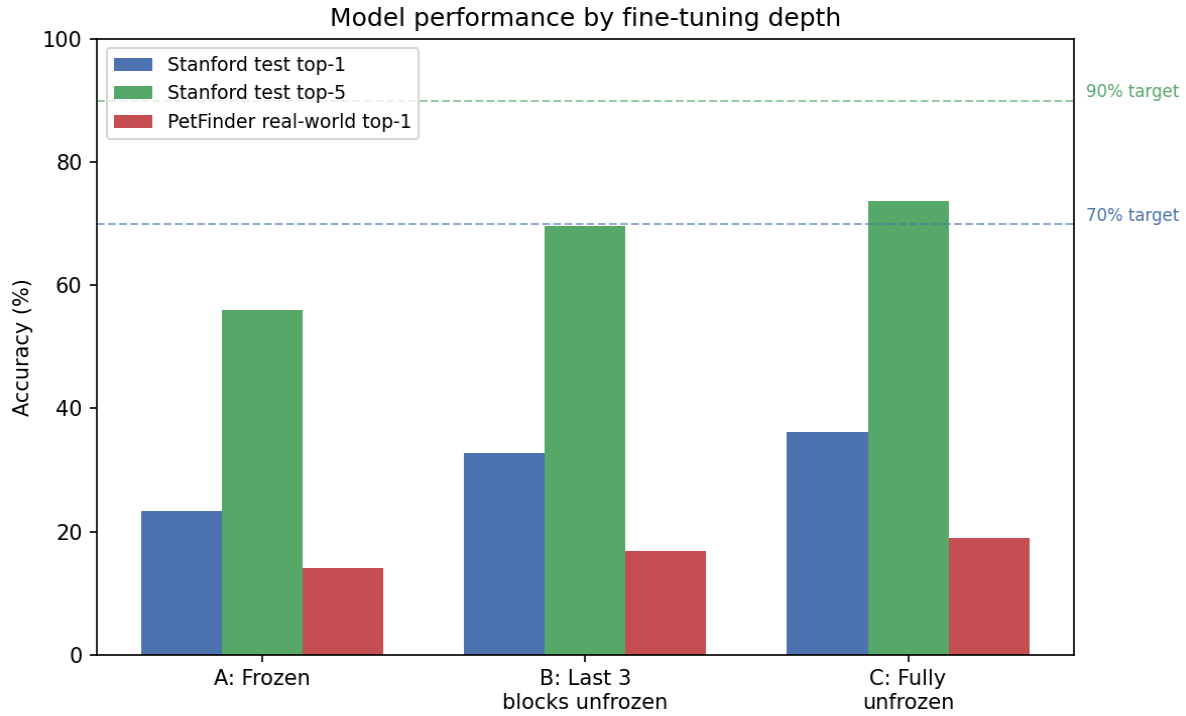


Figure 1. Model performance by fine-tuning depth. Dashed lines mark the original project proposal's target thresholds (70% top-1, 90% top-5).

Model	Stanford top-1	Stanford top-5	Stanford macro F1	PetFinder top-1	PetFinder macro F1
Baseline (stratified)	1.1%	--	0.011	1.1%	0.003
A: Frozen	23.3%	56.0%	0.192	14.0%	0.036
B: Last 3 blocks unfrozen	32.8%	69.6%	0.295	16.9%	0.040
C: Fully unfrozen (final)	36.2%	73.8%	0.340	19.0%	0.048

Table 1. Full model comparison, all metrics.

Against the project proposal's original success criteria ($\geq 70\%$ top-1 / $\geq 90\%$ top-5 on the Stanford test set), the final model falls short -- 36.2% top-1 and 73.8% top-5. Given the model was trained for only 3 epochs on CPU hardware, and every additional degree of fine-tuning freedom improved results with no plateau yet visible, this looks like a training-budget ceiling rather than a ceiling on the approach itself; more epochs on the fully-unfrozen configuration is the most promising next step (see Section 6).

4. Reviewing Predictions Against Actual Outcomes

Per-image predictions on the held-out Stanford test set were compared against true labels to understand where the model succeeds and fails, rather than relying on aggregate accuracy alone.

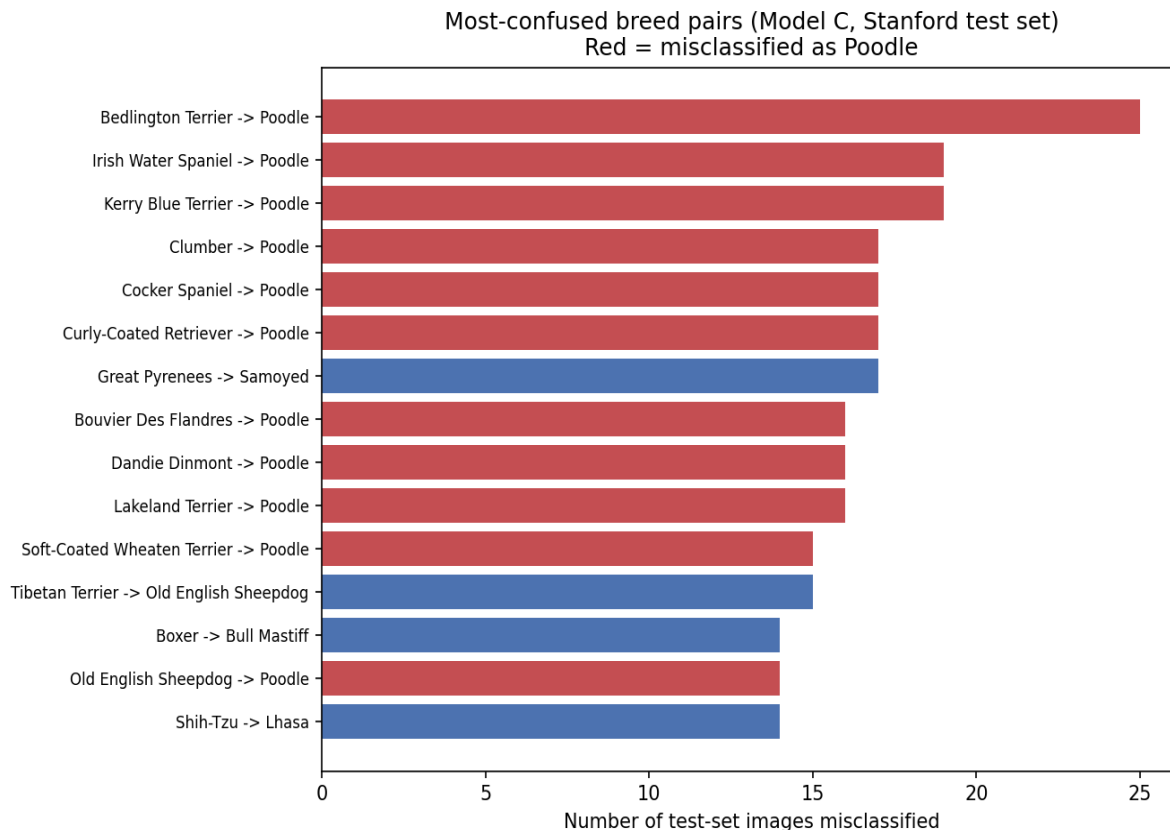


Figure 2. The 15 most common misclassifications on the Stanford test set. Red bars are cases misclassified specifically as 'Poodle'.

The single clearest pattern in the model's errors is a strong bias toward predicting 'Poodle' for many different, unrelated breeds -- 11 of the top 15 confusions funnel into this one class (Bedlington Terrier, Irish Water Spaniel, Kerry Blue Terrier, Clumber, Cocker Spaniel, Curly-Coated Retriever, Bouvier des Flandres, Dandie Dinmont, Lakeland Terrier, Soft-Coated Wheaten Terrier, and Old English Sheepdog were all frequently misclassified as Poodle). Every one of these breeds shares a curly or wavy coat texture with Poodles. This suggests the model, at its current stage of training, has partly learned to key on coat texture as a shortcut rather than the finer-grained features that actually distinguish these breeds -- a plausible and informative failure mode, not random noise.

Notably, this is a *different* finding than the bully-breed misidentification pattern motivating this project (e.g. Pit Bull-type breeds confused with American Staffordshire Terrier). That specific confusion does not appear prominently in the current top-15 errors; the one related case present is Boxer confused with Bull Mastiff, a different but related large-mastiff-type pairing. With only 3 training epochs completed, it is possible this pattern would emerge more clearly with additional training once the coarser coat-texture shortcut is no longer the path of least resistance for the model -- worth specifically re-checking once a longer training run is complete.

5. Real-World Generalization

The final model was also evaluated on PetFinder.my's real, non-staged shelter photos -- a test of whether performance holds up outside Stanford Dogs' cleaner, more posed images.

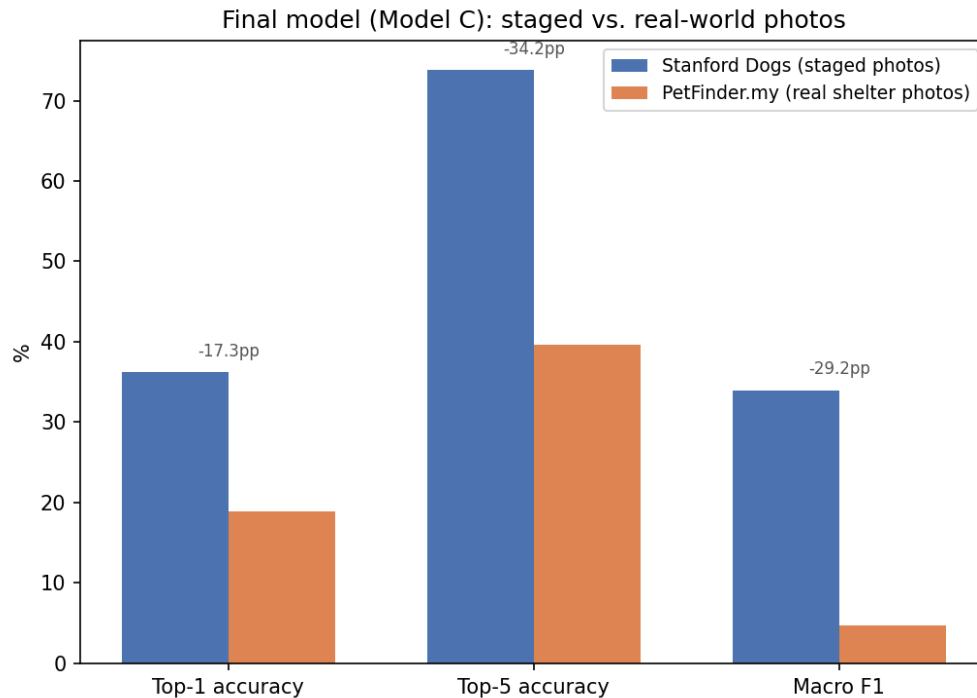


Figure 3. Final model performance: staged Stanford Dogs photos vs. real PetFinder.my shelter photos.

Top-1 accuracy drops from 36.2% to 19.0% moving from staged to real photos -- a meaningful but expected decline given the domain shift the project proposal anticipated. Macro F1 drops far more steeply, from 34.0% to 4.8%, an 86% relative decrease that is proportionally much larger than the accuracy drop. Because macro F1 weights every breed equally regardless of how many examples represent it, this gap indicates the model performs reasonably on the handful of common breeds that dominate PetFinder's matched subset (e.g. Labrador Retriever, Shih-Tzu) while generalizing poorly to the many less-common breeds also present -- confirming the class-imbalance risk flagged in the original proposal, and pointing directly at where future improvement effort matters most: rare-breed performance, not just headline accuracy.

6. Limitations and Further Research

- **Training budget:** only 3 epochs were run on CPU hardware. Since every increase in fine-tuning freedom improved results with no plateau, more epochs on the fully-unfrozen configuration is the single most promising next step.
- **Coat-texture shortcut:** the Poodle-confusion pattern suggests the model may need either more training or targeted data augmentation (e.g. texture-invariant augmentation) to move past coat texture as a primary cue.
- **Real-world rare-breed performance:** the macro-F1 gap on PetFinder points to class imbalance as the main real-world weak point, not domain shift alone. Class-weighted loss or targeted oversampling of rare breeds during training is a natural next experiment.
- **No DNA ground truth:** as noted in the original proposal, all breed labels used here (Stanford Dogs, PetFinder, and any future shelter comparison) are human-assigned, not DNA-verified -- results describe agreement with human labels, not biological accuracy.

7. Recommendations

Based on these findings, three concrete recommendations for a shelter considering a tool like this:

- **Use as a second-opinion flag, not a replacement for staff judgment.** Given current accuracy (36% top-1 on clean photos, 19% on real ones), the model is not reliable enough to assign labels outright. It is reliable enough to flag listings where the model and staff disagree for a second look before publishing -- exactly the role proposed at the outset of this project.
- **Prioritize more training over broader scope.** The clear, unbroken improvement from frozen to fully-unfrozen fine-tuning suggests the fastest path to a usable model is more epochs on the current setup, not a larger architecture or more breeds -- a lower-cost next step than it might first appear.
- **Treat top-5, not top-1, as the practical success measure for now.** At 73.8% top-5 accuracy, the correct breed is very often among the model's top few guesses even when not the single top pick -- a shortlist of likely breeds is a realistic, useful output today, even before top-1 accuracy is strong enough to trust alone.